AUTONOMOUS ENTERPRISE INTELLIGENCE PLATFORM

SHOPPER AI

MACHINE LEARNING + GENERATIVE AI

Project Domain	Machine Learning, RAG and Generative AI
Dataset	Online Shoppers Purchasing Intention
Application	Streamlit web application
Core Capabilities	Analysis • Prediction • Model Evaluation • RAG Chatbot • AI Agent

PROJECT REPORT

Prepared for academic project submission

1. ABSTRACT & INTRODUCTION

ABSTRACT

Shopper AI is an Autonomous Enterprise Intelligence Platform that combines machine learning (ML), retrieval-augmented generation (RAG), and a Streamlit interface to explore online shopping sessions and estimate purchase intention. The project uses the Online Shoppers Purchasing Intention dataset and brings together dataset analysis, model evaluation, purchase prediction, a knowledge-grounded chatbot, AI-assisted insights, and an AI-agent interface.

The dataset analysis recorded 12,330 sessions and 18 columns, with no missing values and 125 duplicate rows. Of these sessions, 1,908 resulted in a purchase (Revenue=True; approximately 15.47%) and 10,422 did not (Revenue=False; approximately 84.53%). The target is imbalanced, so model assessment should consider precision, recall, F1-score, and the confusion matrix rather than accuracy alone.

1. INTRODUCTION

E-commerce websites observe session-level behaviour such as pages visited, time spent on page categories, bounce and exit rates, traffic source, visitor type, and other contextual attributes. Analysing these signals can help identify patterns associated with completed purchases. A classification model can learn from historical sessions and estimate the likelihood that a session belongs to the purchase class.

The project packages data exploration, model outputs, and natural-language question answering in one application so that users can interact with the analysis without manually running every script. It is a prototype for learning and demonstration; predictions should be interpreted as estimates, not guarantees of customer behaviour.

2. PROBLEM, OBJECTIVES & DATASET

2. PROBLEM STATEMENT

The goal is to make session data easier to understand and to provide a consistent workflow for purchase-intention prediction. The system must support both numerical analysis and document-based question answering. These are related but distinct: the ML model predicts from structured, preprocessed features, while RAG answers questions from retrieved text in the knowledge document.

3. OBJECTIVES

- Inspect and summarise the dataset, including dimensions, missing values, duplicates, and target distribution.
- Prepare categorical and numerical features for classification using a consistent preprocessing and encoding workflow.
- Train and compare classification models and save the selected fitted model.
- Review evaluation results using suitable metrics, cross-validation, threshold tuning, and confusion-matrix analysis.
- Provide a Streamlit interface for dataset analysis and purchase prediction.
- Implement a RAG chatbot that retrieves relevant passages from a project knowledge PDF.
- Present AI-assisted insights and an agent page for routing user requests to supported capabilities.

4. DATASET PROFILE

Each row represents a browsing session. The target column Revenue indicates whether the session resulted in a purchase. The analysis reported 12,330 rows, 18 columns, 0 missing values, and 125 duplicate rows. Duplicate rows should be reviewed according to the project's cleaning policy before any final retraining.

Dataset measure	Recorded result
Sessions / rows	**12,330**
Columns	**18**
Missing values	**0**
Duplicate rows	**125**
Purchase sessions	**1,908 (15.47%)**
Non-purchase sessions	**10,422 (84.53%)**

The feature set includes page-visit and page-duration measures, bounce and exit rates, page value, special-day information, operating-system and browser identifiers, region and traffic type, visitor type, weekend status, and Revenue.

3. DATA PREPARATION & ML METHODOLOGY

5. DATA PREPARATION

The project scripts cover preprocessing, categorical encoding, data separation, imbalance handling, model training, tuning, cross-validation, threshold selection, error/outlier analysis, and prediction. The target Revenue is treated as the class to predict. A reproducible workflow should split the data before fitting transformations that learn from the data, then apply the same fitted preprocessing to validation, test, and future prediction inputs.

The saved prediction model was tested with a processed input schema of 29 features. This is the transformed model-input dimension and is not the same as the original dataset's 18 columns. Input feature order, encoding, and transformation must match the model's training pipeline.

6. CLASSIFICATION MODELS

The project documentation identifies the following candidate classifiers:

Model	Role in comparison
Logistic Regression	**Linear baseline classifier**
Decision Tree	**Rule-based nonlinear classifier**
Random Forest	**Ensemble of decision trees**
K-Nearest Neighbours (KNN)	**Distance-based classifier**

Model tuning and cross-validation are used to compare candidates and investigate stability. The selected model artifacts include best_model.joblib and best_tuned_model.joblib. The current report does not invent numeric scores: the exact accuracy, precision, recall, and F1 values must be copied from the latest evaluation CSVs generated by the local project.

7. EVALUATION METRICS

- Accuracy measures the proportion of all predictions that are correct.
- Precision measures how many predicted purchase sessions were actually purchases.
- Recall measures how many actual purchase sessions the model identified.
- F1-score balances precision and recall.
- Confusion matrix summarises true positives, true negatives, false positives, and false negatives.

Since non-purchase sessions are the majority class, a high accuracy score alone may be misleading. The preferred model should be evaluated on held-out data and assessed for its ability to identify the minority purchase class.

4. MODEL RESULTS & PREDICTION

8. MODEL RESULTS AND INTERPRETATION

The project contains a model-training and evaluation workflow with outputs for model evaluation, cross-validation, hyperparameter tuning, threshold tuning, prediction results, and a confusion matrix. These artifacts are intended to support a measured comparison rather than selecting a model by accuracy alone.

Result artifact	What it contributes
evaluation_results.csv	**Recorded metrics for candidate/final model evaluation**
cross_validation_results.csv	**Performance stability across validation folds**
tuning_results.csv	**Results from hyperparameter search**
threshold_results.csv	**Effect of changing the classification threshold**
prediction_results.csv	**Actual and predicted class labels**
confusion_matrix.png	**Visual summary of classification errors**
outlier_report.csv	**IQR-based outlier counts for selected numeric columns**

A threshold of 0.40 is used in one of the project's prediction-evaluation scripts. Threshold choice affects the balance between false positives and false negatives and should be justified from the saved threshold results. The final submission should include the current metric values from the evaluation folder; they cannot be accurately reconstructed from screenshots of the application alone.

8.1 SAMPLE PREDICTION TEST

During a sample prediction-tool test, the saved tuned model loaded successfully and returned a non-purchase prediction with a reported purchase probability of approximately 0.22% for the sample input. This is one example only, not an overall model-performance score. The result is valid only if the sample's 29 processed features match the saved model's expected input schema.

8.2 DATA DISTRIBUTION RESULT

The dataset contains a minority purchase class (15.47%) and a majority non-purchase class (84.53%). This distribution motivates the project's imbalance handling, threshold experiments, and use of class-sensitive metrics.

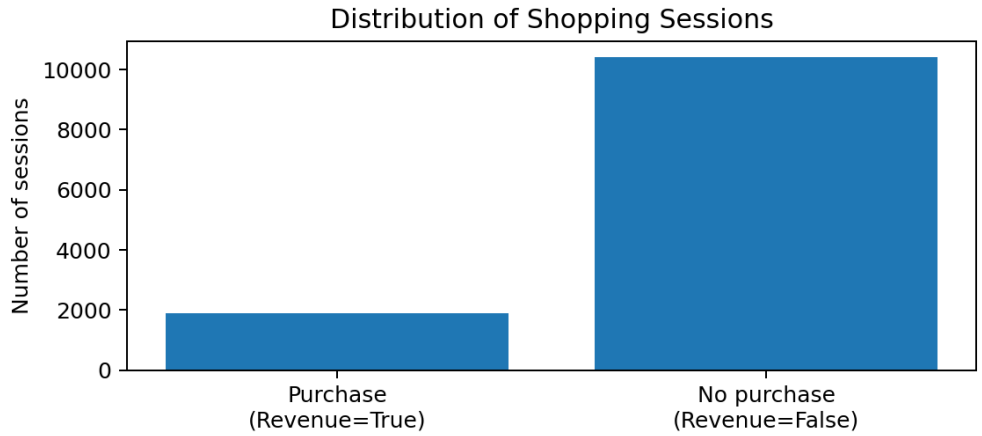


Figure 1. Recorded distribution of purchase and non-purchase sessions.

5. RAG, GENAI & AI AGENT

9. RETRIEVAL-AUGMENTED GENERATION (RAG)

The RAG chatbot answers natural-language questions by searching the project knowledge document, Online_Shoppers_Dataset_Knowledge_Base_5_Pages.pdf. A typical pipeline loads the PDF, splits its text into chunks, creates embeddings, indexes those chunks, retrieves relevant passages for a question, and provides the retrieved context to a language model to generate a response.

In testing, the chatbot accepted a dataset question, displayed a knowledge-base search status, and generated an answer with retrieved PDF sources available in the interface. It correctly returned the dataset dimensions (12,330 rows and 18 columns). For a follow-up question asking how many sessions resulted in a purchase, it reported that the retrieved context did not contain that number. The dataset analysis separately records 1,908 purchase sessions. This demonstrates that RAG response coverage depends on what the document contains and what the retriever returns.

10. GENERATIVE AI INTEGRATION

The application uses a Groq-hosted language model through its configured LangChain integration to produce natural-language answers. The knowledge retrieval step is intended to ground responses in project material. API credentials should be kept in a local .env file or environment variable and must never be published in source control or included in the report.

11. AI AGENT AND TOOL ROUTING

The AI Agent & Tools page provides an interface for interpreting user requests and routing supported requests to capabilities such as RAG question answering and dataset analysis. The project also contains a prediction tool and a separate Purchase Prediction page. The report describes the agent according to the demonstrated interface; it does not claim that every tool is autonomously executed by the agent if the current implementation instead directs the user to a separate page.

12. SYSTEM ARCHITECTURE

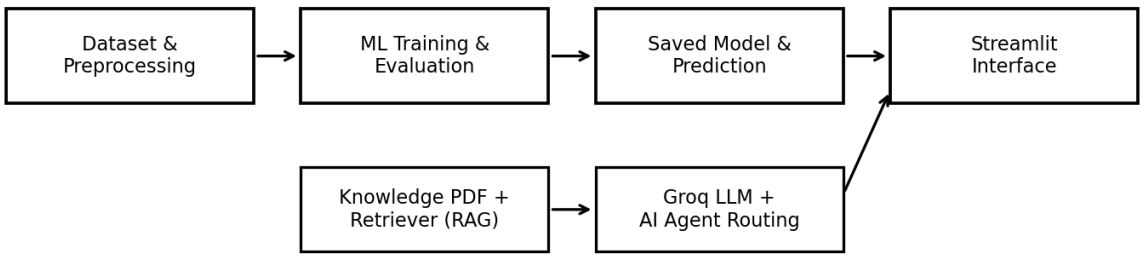


Figure 2. High-level view of the platform components and their relationships.

6. STREAMLIT APP, TESTING & CONCLUSION

13. STREAMLIT APPLICATION MODULES

Module	Purpose
Home Dashboard	**Project overview and key indicators**
Dataset Analysis	**Dataset summary and quality checks**
RAG Chatbot	**Question answering grounded in the project PDF**
Purchase Prediction	**Prediction from model-compatible features**
ML Model Comparison	**Displays recorded model evaluation metrics**
AI Insights & Recommendations	**Explains observed statistics and suggests hypotheses to test**
Evaluation & Reliability	**Shows saved evaluation outputs and visualisations**
AI Agent & Tools	**Demonstrates routing to supported capabilities**
About Project	**Project summary and technology information**

14. TESTING SUMMARY

- Dataset analysis displayed 12,330 rows, 18 columns, no missing values, and 125 duplicate rows.
- Purchase class distribution was reported as 1,908 purchases and 10,422 non-purchases.
- RAG accepted a question and generated a response; it answered the dataset dimensions but did not retrieve the purchase-session count in the tested query.
- The saved prediction model loaded and generated a sample prediction. Correct feature encoding and ordering remain essential.
- Evaluation outputs are saved as CSV reports and visualisations and should be checked against the latest run before final submission.

15. LIMITATIONS

- The model's exact final performance metrics must be taken from the latest evaluation files.
- Class imbalance can reduce the usefulness of accuracy as a standalone metric.
- RAG may not answer a question if the knowledge document lacks the relevant fact or retrieval misses it.
- The prediction workflow depends on compatible preprocessed features.
- The current agent interface should only be described as autonomous for tools it actually invokes.

16. CONCLUSION AND FUTURE WORK

Shopper AI integrates structured dataset analysis, ML-based purchase prediction, model-evaluation artefacts, a RAG chatbot, and AI-agent routing in a single Streamlit application. The work demonstrates how ML and generative AI can complement each other: ML estimates a class from structured features, while RAG helps users ask questions about dataset concepts in natural language.

Future work includes adding verified dataset statistics to the knowledge PDF and rebuilding the retrieval index, reporting final model metrics with the test methodology, improving prediction input validation, strengthening model explainability, and extending agent routing with clearer tool execution and error handling.

REFERENCES

- Online Shoppers Purchasing Intention dataset used in the local project.
- Online_Shoppers_Dataset_Knowledge_Base_5_Pages.pdf (project knowledge source).
- Scikit-learn documentation — https://scikit-learn.org/stable/
- Streamlit documentation — https://docs.streamlit.io/
- LangChain documentation — https://python.langchain.com/

Prepared from project details and observed tests available during the development conversation. Add student, college, guide, and submission details before printing if required.